

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report:
Configure Authentication and Authorization in Linux

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Background / Scenario

In this lab, you will use the Linux command line to create a group for new users and add users to the group. Each user will be assigned a password for authenticating at login. Then you will modify permissions to authorize read, write, and execute privileges for both users and groups.

Required Resources

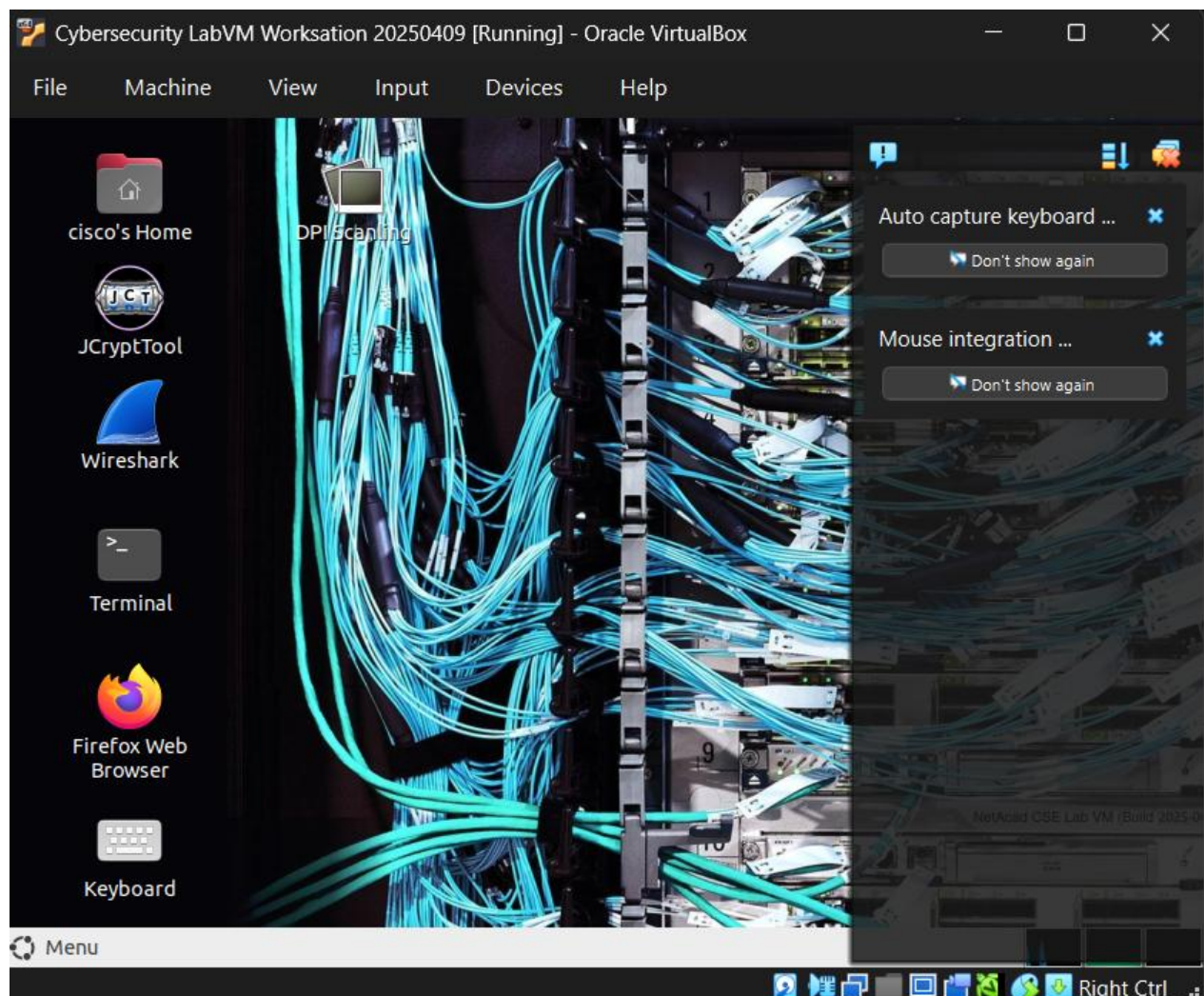
- PC with the CSE-LABVM installed in VirtualBox

Part 1: Add a New Group for Users

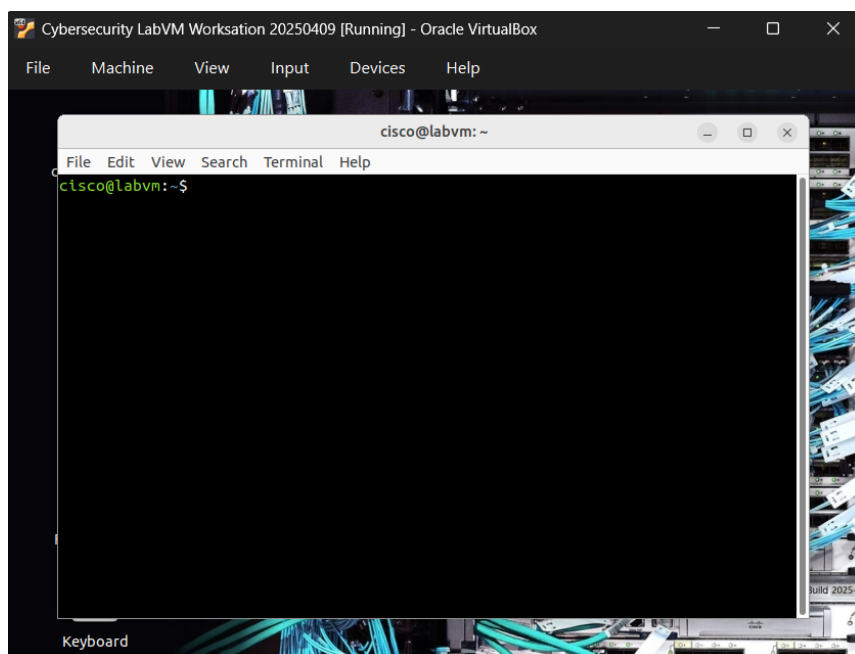
In this part, you will add a new group for users to the virtual machine.

Step 1: Open a terminal window in the CSE-LABVM.

- a. Launch the CSE-LABVM.

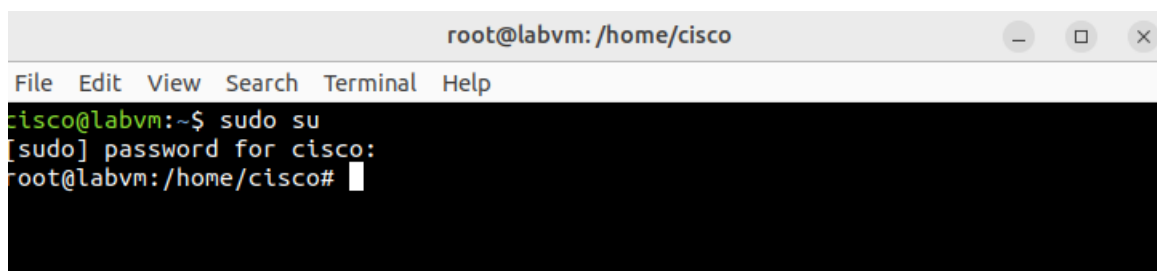


- b. Double-click the Terminal icon to open a terminal.



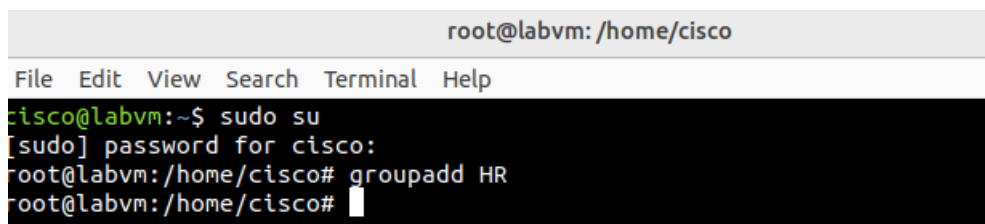
Step 2: Escalate privileges to the root level.

Enter the `sudo su` command and enter password as the password when prompted.



Step 3: Add a new group named HR.

Enter the `groupadd HR` command.



Step 4: Verify the new group has been added.

Enter the `cat /etc/group` command to verify that HR was added.

```
root@labvm:/home/cisco# cat /etc/group
root@labvm:/home/cisco
File Edit View Search Terminal Help
nopasswdlogin:x:119:cisco
netdev:x:120:
pulse:x:121:
pulse-access:x:122:
cisco:x:1001:
analyst:x:1002:
sec_admin:x:1003:
vboxsf:x:999:cisco,analyst,sec_admin
vboxdrmpc:x:998:
wireshark:x:123:cisco,analyst,sec_admin
telnetd:x:124:
Alice:x:1004:
Bob:x:1005:
Eve:x:1006:
Eric:x:1007:
Xnobody:x:1008:
snort:x:125:
ssl-cert:x:126:freerad
ntp:x:127:
ftp:x:128:
tftp:x:129:
freerad:x:130:
HR:x:1009:
root@labvm:/home/cisco#
```

The new group HR is shown at the bottom of the `/etc/group` file with a group ID of 1005.

Part 2: Add Users to the New Group

In this part, you will add user accounts for Jenny and Joe to the HR group.

Step 1: Add Jenny as a new user and move her to the HR group.

- a. Complete the following to add Jenny as a user:
 - 1) Enter the `adduser jenny` command and press Enter.
 - 2) Enter `jenPass` as the password, and then press Enter.
 - 3) Re-type the new password, and then press Enter.
 - 4) Enter Jenny for Full Name, and then press Enter.
 - 5) For the rest of the configuration, press Enter.
 - 6) Enter Y to verify the information is correct, and then press Enter.

```
root@labvm:/home/cisco# adduser jenny
Adding user `jenny' ...
Adding new group `jenny' (1000) ...
Adding new user `jenny' (1000) with group `jenny' ...
Creating home directory `/home/jenny' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for jenny
Enter the new value, or press ENTER for the default
    Full Name []: Jenny
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] Y
root@labvm:/home/cisco#
```

- b. Move jenny to the HR group. Enter the `usermod -G HR jenny` command.

```
root@ubuntu:/home/cisco# usermod -G HR jenny
```

Step 2: Add Joe as a new user and move him to the HR group.

- a. Enter the adduser joe command and complete the steps to assign joe the password joePass and full name Joe.

```
root@labvm:/home/cisco# adduser joe
Adding user `joe' ...
Adding new group `joe' (1010) ...
Adding new user `joe' (1009) with group `joe' ...
Creating home directory `/home/joe' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for joe
Enter the new value, or press ENTER for the default
    Full Name []: Joe
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] Y
root@labvm:/home/cisco#
```

- b. Place the user joe in the HR group.

```
root@labvm:/home/cisco# usermod -G HR joe
```

Step 3: Verify the newly created users in the passwd file.

Enter the cat /etc/passwd command to verify the new users in the passwd file.

```
root@labvm:/home/cisco# cat /etc/passwd
uidd:x:108:114::/run/uidd:/usr/sbin/nologin
tcpdump:x:109:115::/nonexistent:/usr/sbin/nologin
tss:x:110:116:TPM software stack,,,:/var/lib/tpm:/bin/false
landscape:x:111:117::/var/lib/landscape:/usr/sbin/nologin
usbmux:x:112:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
lxd:x:999:100::/var/snap/lxd/common/lxd:/bin/false
lightdm:x:113:118:Light Display Manager:/var/lib/lightdm:/bin/false
pulse:x:114:121:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
cisco:x:1001:1001:1. Cybersecurity Analyst - Support Technician (Cybersecurity Essentials):/home/cisco:/bin/bash
analyst:x:1002:1002:2. SOC Analyst (CyberOps Associate):/home/analyst:/bin/bash
sec_admin:x:1003:1003:3. Security Engineer (Network Security):/home/sec_admin:/bin/bash
vboxadd:x:998:1::/var/run/vboxadd:/bin/false
telnetd:x:115:124::/nonexistent:/usr/sbin/nologin
Alice:x:1004:1004::/home/Alice:/bin/bash
Bob:x:1005:1005::/home/Bob:/bin/bash
Eve:x:1006:1006::/home/Eve:/bin/bash
Eric:x:1007:1007::/home/Eric:/bin/bash
Xnobody:x:1008:1008::/home/Xnobody:/bin/sh
snort:x:116:125:Snort IDS:/var/log/snort:/usr/sbin/nologin
ntp:x:117:127::/nonexistent:/usr/sbin/nologin
ftp:x:118:128:ftp daemon,,,:/srv/ftp:/usr/sbin/nologin
tftp:x:119:129:tftp daemon,,,:/srv/tftp:/usr/sbin/nologin
freerad:x:120:130::/etc/freeradius:/usr/sbin/nologin
speech-dispatcher:x:121:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
jenny:x:1000:1000:Jenny,,,:/home/jenny:/bin/bash
joe:x:1009:1010:Joe,,,:/home/joe:/bin/bash
root@labvm:/home/cisco#
```

Step 4: View the created users in the shadow file.

Enter the `cat /etc/shadow` command to verify the new users in the shadow file.

```
root@labvm: /home/cisco# cat /etc/shadow
systemd-network*:19213:0:99999:7:::
systemd-resolve*:19213:0:99999:7:::
messagebus*:19213:0:99999:7:::
systemd-timesync*:19213:0:99999:7:::
pollinate*:19213:0:99999:7:::
sshd*:19213:0:99999:7:::
syslog*:19213:0:99999:7:::
uuidd*:19213:0:99999:7:::
tcpdump*:19213:0:99999:7:::
tss*:19213:0:99999:7:::
landscape*:19213:0:99999:7:::
usbmux*:19398:0:99999:7:::
lxd!:19398:::
lightdm*:19398:0:99999:7:::
pulse*:19398:0:99999:7:::
cisco:$6$37i1ivfBmMzNZhFC$HXKTrMgxqc/J4IW.5jCK01GeCWfKGLxbpLVvOXZTTDjQEKQhIFl1iaU0s6m.CjfnC8guZdZbqDscHyDUymtF0:19403:0:99999:7:::
analyst:$6$tWdBie2bEosRDjV$XWHL5c8eAEZq/zEwKkzbkcE86vZso6tx0kyrVvaUjg16AKZNRllFP49wDfHJrXwI.aRTQL/.Tf1b6XNAhs4Ei0:19403:0:99999:7:::
sec_admin:$6$QmW70kc28oxqIYN$DxF5oMeMG2tjQ5usv.Ivt2Cdra20oIueeFhD5xWXW6PGUUXfJwLVI6lcpzcW4VA5h1KaMVSUjTI2u5WgI/Xe1:19403:0:99999:7:::
vboxadd!:19398:::
telnetd*:19398:0:99999:7:::
Alice:$6$0MvdeefbBpxgZWCF$DJ/Lp.npniPymCL5xU/P7/wsOMqJtj0tSFUmuqhDTa.SJhs/Qgzo6QQ2bV9RP/UB01giBdiAAySvVoW0XFtS/.:19403:0:99999:7:::
Bob:$6$A7RobK0k/jmAMjYd$P005EE.R1cohiE2mOMegJB9uUEKH8z8fXCsxUfPtYkpqWrs8lk5yHueqnFT6PatbGg0BeH/.bqGPNL7ffgfm/:19403:0:99999:7:::
Eve:$6$dxM39h24p9vri1Hx$ah.gVt4fd0Ej3BYKcAGFEpowk/L5vvNcxFNLLd7l6JdPQhnK0ImKcjiw/epEB9sXbBxRYBLx20iRviQkNCZTi0:19403:0:99999:7:::
Eric:$6$2uif2qGy6blp3WeZ$AgebBtU1owduQns5MSa3BmKxFPLQaVIZEafS2.f9KB2rRQHMxIr7AQrDnZqcgAKdOHmIREHrsNqYkTqzQq1Zv/:19403:0:99999:7:::
Xnobody!:19398:0:99999:7:::
snort*:19398:0:99999:7:::
ntp*:19398:0:99999:7:::
ftp*:19398:0:99999:7:::
tftp*:19398:0:99999:7:::
freerad*:19398:0:99999:7:::
speech-dispatcher!:20187:0:99999:7:::
jenny:$6$2hgnmuF0mgTauwlc$juG1Jb8rFEE71UIoxMUSNZScGmbyQPN0P6H1dfHChKc9Fjq50Q3MOTvyKMypnANj0jGCV/BQtyMKCXc17sadD0:20633:0:99999:7:::
joe:$6$Q0n5b$JN0zpaHEH/$dqpyYcxz3c5eZsZolHrqJgsnbjCYKLH.GTNmCnshbfJadA3Co8nRBGuVvraeWBwRzcu.WsyB3i4rQdYcjyCt1:20633:0:99999:7:::
root@labvm: /home/cisco#
```

Part 3: Switch Users and Modify Permissions

In this part, you will login as jenny, explore directories, and change permissions.

Step 1: Switch user from root to jenny.

- To switch to Jenny's desktop, click Menu at the bottom left of the desktop, and then click Logout.
- Click Switch User from the dialog box.
- Click Jenny from the list of available users and enter the password jenPass.
- Jenny's desktop loads. Right-click the desktop and choose Open in Terminal.

```
jenny@labvm:~/Desktop$
```

Step 2: Explore Jenny's environment.

- Enter the pwd command to print the current directory, then navigate to /home with cd ../../ command.

```
jenny@labvm:~/Desktop$ pwd
/home/jenny/Desktop
jenny@labvm:~/Desktop$ cd ../../
jenny@labvm:/home$
```

- Enter the ls -l command to list all directories in /home, their permissions, and their users.

```
jenny@labvm:/home$ ls -l
total 28
drwxr-xr-x  2 Alice  Alice  4096 Mar 18 21:58 Alice
drwxr-xr-x  2 Bob    Bob    4096 Mar 18 21:58 Bob
drwxr-xr-x 12 cisco  cisco  4096 Mar 19 20:02 cisco
drwxr-xr-x  2 Eric   Eric   4096 Mar 18 21:58 Eric
drwxr-xr-x  2 Eve    Eve    4096 Mar 18 21:58 Eve
drwxr-xr-x  9 jenny  jenny  4096 Mar 19 19:58 jenny
drwxr-xr-x  2 joe    joe    4096 Mar 19 19:44 joe
```

The Linux permission string has 10 characters. For example: drwxr-xr-x

Field	Meaning
d	First character — d = directory, - = file
rwX (positions 2-4)	User/Owner permissions — read, write, execute
r-x (positions 5-7)	Group permissions — read and execute only (no write)
r-x (positions 8-10)	Others permissions — read and execute only (no write)

- As Jenny, enter cd joe to navigate into Joe's directory. This works because the "others" permission includes x (execute), which allows entering a directory.

```
jenny@labvm:/home$ cd joe
jenny@labvm:/home/joe$
```

- d. While in Joe's directory, enter the `touch new.txt` command to create a file. Permission is denied because jenny does not have write access to Joe's directory.

```
jenny@labvm:/home/joe$ touch new.txt
touch: cannot touch 'new.txt': Permission denied
jenny@labvm:/home/joe$
```

- e. Enter the `cd ..` command to exit Joe's home directory.

```
jenny@labvm:/home/joe$ cd ..
jenny@labvm:/home$
```

Step 3: Login as root.

Joe may not want Jenny to read the contents of his directory. Root can change the directory permissions to deny Jenny read access.

- a. Login as user cisco with the `su cisco` command.

```
jenny@labvm:/home$ su cisco
Password:
```

- b. Enter the `sudo -i` command to switch to root.

```
cisco@labvm:~$ sudo -i
[sudo] password for cisco: password
```

Step 4: Modify the permissions for Joe's home directory.

Navigate to `/home` and enter `chmod o-x joe` to remove execute permission for "others" on Joe's directory.

```
root@labvm:~# cd /home
root@labvm:/home# chmod o-x joe
root@labvm:/home# ls -l
total 28
drwxr-xr-x  2 Alice  Alice  4096 Mar 18 21:58
drwxr-xr-x  2 Bob    Bob    4096 Mar 18 21:58
drwxr-xr-x 12 cisco  cisco  4096 Mar 19 20:02
drwxr-xr-x  2 Eric   Eric   4096 Mar 18 21:58
drwxr-xr-x  2 Eve    Eve    4096 Mar 18 21:58
drwxr-xr-x  9 jenny  jenny  4096 Mar 20 14:02
drwxr-xr--  2 joe    joe    4096 Mar 19 19:44
```

Step 5: Verify Jenny can no longer access Joe's directory.

- a. Logout as root and the user cisco.

```
root@labvm:/home# exit
logout
cisco@labvm:~$ exit
logout
jenny@labvm:/home$
```

- b. Enter the `cd joe` command to attempt to navigate to Joe's directory. Permission is now denied.

```
jenny@labvm:/home$ cd joe
bash: cd: joe: Permission denied
jenny@labvm:/home$
```

The table below shows examples of how the `chmod` command can be used in symbolic mode:

Command	Operator Meaning	Effect
<code>chmod u+x file</code>	u = user, + = add, x = execute	Add execute for the file owner
<code>chmod u-x file</code>	u = user, - = remove, x = execute	Remove execute from the file owner
<code>chmod g+w file</code>	g = group, + = add, w = write	Add write permission for the group
<code>chmod g-r file</code>	g = group, - = remove, r = read	Remove read from the group
<code>chmod o+r file</code>	o = others, + = add, r = read	Add read permission for others
<code>chmod o-x file</code>	o = others, - = remove, x = execute	Remove execute from others
<code>chmod a+r file</code>	a = all (u+g+o), + = add, r = read	Add read for everyone
<code>chmod u=rwx file</code>	u = user, = = set exactly	Set user permissions to exactly rwx

Part 4: Modify Permissions in Absolute Mode

In the previous part, you changed permissions in symbolic mode. In this part, you will use the `chmod` command with octal values to set permissions for each triplet (rwx) for user, group, and other.

Step 1: Switch user from jenny to joe.

- Click Menu > End the current session.
- Click Switch User from the dialog box.
- Click Joe from the list of available users and enter the password `joePass`.
- Joe's desktop loads. Right-click the desktop and choose Open in Terminal.

```
joe@labvm:~/Desktop$
```

Step 2: Explore Joe's environment.

- Enter the `pwd` command, then navigate to `/home` with `cd ../../`.

```
joe@labvm:~/Desktop$ pwd
/home/joe/Desktop
joe@labvm:~/Desktop$ cd ../../
joe@labvm:/home$
```

- Enter `ls -l` to list all directories. Notice Joe's directory shows `drwxr-xr--` — others cannot access it.

```
joe@labvm:/home$ ls -l
total 28
drwxr-xr-x  2 Alice Alice 4096 Mar 18 21:58 Alice
```

```
drwxr-xr-x  2 Bob   Bob   4096 Mar 18 21:58 Bob
drwxr-xr-x 12 cisco cisco 4096 Mar 19 20:02 cisco
drwxr-xr-x  2 Eric  Eric  4096 Mar 18 21:58 Eric
drwxr-xr-x  2 Eve   Eve   4096 Mar 18 21:58 Eve
drwxr-xr-x  9 jenny jenny 4096 Mar 20 14:02 jenny
drwxr-xr--  9 joe   joe   4096 Mar 20 15:01 joe
```

Step 3: Use absolute mode to modify and verify Joe's directory permissions.

Absolute (octal) mode uses a three-digit number to represent permissions for owner, group, and other. Each digit is the sum of the permission values below:

Octal Value	Binary	Permission
0	000	--- (no permissions)
1	001	--x (execute only)
2	010	-w- (write only)
3	011	-wx (write and execute)
4	100	r-- (read only)
5	101	r-x (read and execute)
6	110	rw- (read and write)
7	111	rxw (read, write, and execute)

Example — the command `chmod 764 examplefile` sets the following permissions:

	Owner (User)	Group	Others
Octal digit	7	6	4
Permission	rxw	rw-	r--

- a. Modify the "others" field for Joe's folder so others can read and execute but NOT write, while keeping the "user" field as read, write, and execute.

```
joe@labvm:/home$ chmod 705 joe
```

- b. List the file permissions of the current directory to verify the changes.

```
joe@labvm:/home$ ls -l
total 28
drwxr-xr-x  2 Alice Alice 4096 Mar 18 21:58 Alice
drwxr-xr-x  2 Bob   Bob   4096 Mar 18 21:58 Bob
drwxr-xr-x 12 cisco cisco 4096 Mar 19 20:02 cisco
drwxr-xr-x  2 Eric  Eric  4096 Mar 18 21:58 Eric
drwxr-xr-x  2 Eve   Eve   4096 Mar 18 21:58 Eve
drwxr-xr-x  9 jenny jenny 4096 Mar 20 14:02 jenny
drwx---r-x  9 joe   joe   4096 Mar 20 15:01 joe
```

Step 4: Create a file in the joe directory.

Switch to the joe directory, use touch test.txt to create a file, and then list directory contents.

```
joe@labvm:/home$ cd joe
joe@labvm:~$ touch test.txt
joe@labvm:~$ ls -l
total 12
drwxr-xr-x 2 joe joe 4096 Mar 20 15:00 Desktop
drwxr-xr-x 2 joe joe 4096 Mar 20 15:00 Documents
drwxr-xr-x 2 joe joe 4096 Mar 20 15:00 Downloads
-rw-rw-r-- 1 joe joe    0 Mar 20 15:33 test.txt
```

Step 5: Switch user from joe to jenny.

- Click Menu > End the current session.
- Click Switch User from the dialog box.
- Click Jenny from the list and enter the password jenPass.
- Jenny's desktop loads. Open a terminal.

```
jenny@labvm:~/Desktop$
```

Step 6: Change to the home directory and list its contents.

```
jenny@labvm:~/Desktop$ cd ../../
jenny@labvm:/home$ ls -l
total 28
drwxr-xr-x  2 Alice Alice 4096 Mar 18 21:58 Alice
drwxr-xr-x  2 Bob   Bob   4096 Mar 18 21:58 Bob
drwxr-xr-x 12 cisco cisco 4096 Mar 19 20:02 cisco
drwxr-xr-x  2 Eric  Eric  4096 Mar 18 21:58 Eric
drwxr-xr-x  2 Eve   Eve   4096 Mar 18 21:58 Eve
drwxr-xr-x  9 jenny jenny 4096 Mar 20 14:02 jenny
drwx---r-x  9 joe   joe   4096 Mar 20 15:01 joe
```

Step 7: Change to the joe directory and list its contents.

As a member of "others", jenny now has read access to Joe's directory and can read test.txt.

```
jenny@labvm:/home$ cd joe
jenny@labvm:/home/joe$ ls -l
total 12
drwxr-xr-x 2 joe joe 4096 Mar 20 15:00 Desktop
drwxr-xr-x 2 joe joe 4096 Mar 20 15:00 Documents
drwxr-xr-x 2 joe joe 4096 Mar 20 15:00 Downloads
-rw-rw-r-- 1 joe joe    0 Mar 22 14:33 test.txt
```

Step 8: Attempt to create a file in the joe directory.

Jenny is denied permission to write to Joe's directory because the group field is --- (no permissions for group).

```
jenny@labvm:/home/joe$ touch jenny.txt
touch: cannot touch 'jenny.txt': Permission denied
jenny@labvm:/home/joe$
```

Step 9: Switch user from jenny to cisco and close the VM.

- a. Click Menu > End the current session.
 - b. Click Switch User from the dialog box.
 - c. Click Cybersecurity Analyst from the list and enter password as the password.
 - d. Click File > Close, choose Save the machine state, and then click OK.
-